1. Which type of voltage does LM 35 temperature sensor generate?

a. Analog
b. Digital
c. Both analog and digital
d. It is unable to generate any voltage

Answer: (a)

Justification: Please refer to Slide 15 of Lecture 1 of Week 6

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2. Temperature sensor DS1621 is a _____ sensor, which generates _____ bits temperature data.

a. analog, 5
b. analog, 9
c. digital, 5
d. digital, 9

Answer: (d)

Justification: Please refer to Slide 16 of Lecture 1 of Week 6

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3. Which of the following is the application of Ultrasonic sensor?

a. Detects radiations coming from human body in its surrounding area
b. Distance measurement, pattern matching, color checking, structured lighting, and motion capture
c. Automatic door open/close, human detection, lift lobby, common staircase, and shopping Mall
d. Liquid level monitoring of tank, trash level monitoring

Answer: (d)

Justification: Please refer to Slide 25 of Lecture 1 of Week 6

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4. What is the Baseline Resistance of a sensor?

a. Resistance of the sensor material in air when not exposed to the target gas
b. Resistance of the sensor material in air when exposed to the target gas
c. Resistance of the sensor material when not exposed to the atmosphere
d. Resistance of the sensor material when exposed to the atmosphere

Answer: (a)

Justification: Please refer to Slide 4 of Lecture 2 of Week 6

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5. Which type of sensor decreases its resistance w.r.t. the baseline resistance when exposed to a reducing gas?

a. c-type sensors
b. p-type sensors
c. n-type sensors
d. All the sensor types

Answer: (c)

Justification: Please refer to Slide 5 of Lecture 2 of Week 6

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6. Define the Sensitivity of a gas sensor.

a. It is the ability to detect a particular gas in a mixture of different gases.
b. It is the change in the output signal with respect to unit change in input.
c. It is characteristic of the sensor which ensures its resistance to return to its base resistance value if exposure to the target gas is stopped.
d. None of the above

Answer: (b)

Justification: Please refer to Slide 6, 7 of Lecture 2 of Week 6

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7. Which of the following protocol falls under the category of Industrial Ethernet??

a. CC-Link
b. DeviceNet
c. Time-Sensitive Networking (TSN)
d. Interbus

Answer: (c)

Justification: Please refer to Slide 3 of Lecture 3 of Week 6

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8. What are the two units defined in the ModBus-TCP data frame?

a. Protocol Data Unit and Application Data Unit
b. Protocol Data Unit and Communication Data Unit
c. Communication Data Unit and Application Data Unit
d. None of the above

Answer: (a)

Justification: Please refer to Slide 6 of Lecture 3 of Week 6

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9. Which of the following network topology/(ies) is/are not supported by EtherCat?

a. tree
b. hybrid
c. star
d. mesh

Answer: (d)

Justification: Please refer to Slide 11 of Lecture 3 of Week 6

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10. What is the maximum data rate per packet of EtherNet/IP?

a. 500 bytes
b. 100 bytes
c. 1500 bytes
d. 2000 bytes

Answer: (c)

Justification: Please refer to Slide 15 of Lecture 3 of Week 6

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11. What is the full form of Profinet?

a. PReplanned FIeld NETwork

b. PROcess Figured NETwork
c. PReplanned Figured NETwork
d. PROcess FIeld NETwork

Answer: (d)

Justification: Please refer to Slide 17 of Lecture 3 of Week 6

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12. Which of the following communication infrastructure is an example of wired connectivity?

a. LPWAN
b. PSTN
c. IEC-PAS 62601/WIA-PA
d. ISA 100

Answer: (b)

Justification: Please refer to Slide 16 of Lecture 4 of Week 6

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13. Which of the following is/are the limitation(s) of satellite communication?

a. Launching of satellite in space comes at a higher cost
b. Propagation delay is more compared to other terrestrial methods
c. Difficulty in repairs in case of any damage
d. All of the above

Answer: (d)

Justification: Please refer to Slides 28 of Lecture 4 of Week 6

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14. What is the frequency spectrum of mmWave for cellular communication?

a. 30 – 100 GHz
b. 30 – 100 MHz
c. 57 – 64 GHz
d. 57 – 64 MHz

Answer: (a)

Justification: Please refer to Slide 26 of Lecture 5 of Week 6

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15. The output voltage of LM 35 temperature sensor is _____ proportional to the Celsius temperature.

a. quadratically
b. exponential
c. linearly
d. None of the above

Answer: (c)

Justification: Please refer to Slide 15 of Lecture 1 of Week 6

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